Bearings

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Bearings Basics

1. 000°

2. 090°

3. 008°

4. C) 240

5. East

6. 180°

7. C) South

8. N, NE, E, SE, S, SW, W, NW

9. 045°

10. D) North-West

11. A, C and D are valid (B is not written with three digits; E exceeds a full turn of 360°)

12. 270°

13. They are exact opposites — 090° and 270° differ by 180°, so each is the back bearing of the other

14. B and C (150° and 200° lie between 090° and 270°; 060°, 300° and 350° do not)

15. A) North

Reading & Finding Bearings

16. North-East (a bearing of 050° lies between North, 000°, and East, 090°, closer to North-East, 045°)

17. 245° (065° + 180° = 245°)

18. 322° (142° + 180° = 322°)

19. C) 120° (300° − 180° = 120°)

20. 030° (210° − 180° = 030°)

21. A, C and D are true (B is false — 095° + 180° = 275°, not 185°; E is false — 300° − 180° = 120°, not 110°)

22. 75° (turning clockwise from North through 75° gives a bearing of 075°)

23. B) South-East (128° lies between East, 090°, and South, 180°, closer to South-East, 135°)

24. 010° (190° − 180° = 010°)

25. The back bearing points roughly South-West; adding 180° to 048° gives a back bearing of 228°

26. B) 178° (358° is 180° or more, so subtract 180°: 358° − 180° = 178°)

27. A, C and E are true (B is false — the back bearing is found by ADDING 180° when the bearing is less than 180°, not subtracting it; D is false — 265° is greater than 180°, not less)

28. 032° (212° − 180° = 032°)

29. 315° (North-West is a bearing of 315°)

30. 130° (310° − 180° = 130°)

Bearings Challenge

31. 330° (the B-to-C leg was on a bearing of 150°, so retracing it from C to B uses the back bearing: 150° + 180° = 330°)

32. 252° (072° + 180° = 252°)

33. 308° (128° + 180° = 308°)

34. C) 120° (160° − 040° = 120°)

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35. 185° (095° + 90° = 185°)

36. A, B, D and E are true (C is false — since 162° is less than 180°, the back bearing is found by ADDING 180°, not subtracting it)

37. Corrected bearing = 220° (205° + 15°); back bearing = 040° (220° − 180°)

38. A) 90° (the angle between bearings 030° and 300°, measured the short way round, is 030° + (360° − 300°) = 030° + 60° = 90°)

39. C is the same point as A — the back bearing of 085° is 085° + 180° = 265°, so the B-to-C leg exactly retraces the A-to-B leg in reverse, bringing the plane back to its start

40. 130° (310° − 180° = 130°)

41. A, B, C and E are true (D is false — 180° − 078° = 102°, which is not the back bearing; the correct back bearing is 078° + 180° = 258°)

42. The ship's return bearing is 200° − 180° = 020°. Since 020° and 200° differ by exactly 180°, they point in exactly opposite directions, so the second ship sailing out on 020° travels the exact opposite way to the first ship's original outward journey on 200°

43. 322° (142° + 180° = 322°); this is closest to D) North-West (315°)

44. Since 265° is exactly the back bearing of 085° (085° + 180° = 265°), the B-to-C leg lies on the same straight line as the A-to-B leg, heading back towards A. As B-to-C is 5 km (shorter than the 12 km A-to-B leg), C is 12 − 5 = 7 km from A. Direct distance A to C = 7 km; via B the ship travelled 12 + 5 = 17 km; so sailing directly would have saved 17 − 7 = 10 km

45. x + 200 = 2x + 180, so 200 − 180 = 2x − x, giving x = 20 (outward bearing = 2(20) = 40°, back bearing = 20 + 200 = 220°, and 40° + 180° = 220° ✓)

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Homework Worksheet — Answers

Bearings Basics

1. 180°

2. 270°

3. 005°

4. C) 075

5. North

6. 000°

7. D) West

8. N, NW, W, SW, S, SE, E, NE

9. 225°

10. A) North-East

11. A, C and E are valid (B is not written with three digits; D exceeds a full turn of 360°)

12. 090°

13. They are exact opposites — 000° and 180° differ by 180°, so each is the back bearing of the other

14. A, D and E (030°, 320° and 355° are less than 090° or greater than 270°; 100° and 200° are not)

15. C) South

Reading & Finding Bearings

16. South-West (a bearing of 220° lies between South, 180°, and West, 270°, closer to South-West, 225°)

17. 215° (035° + 180° = 215°)

18. 018° (198° − 180° = 018°)

19. C) 260° (080° + 180° = 260°)

20. 335° (155° + 180° = 335°)

21. A, C and D are true (B is false — 100° + 180° = 280°, not 190°; E is false — 060° + 180° = 240°, not 260°)

22. 112° (turning clockwise from North through 112° gives a bearing of 112°)

23. C) North-West (302° lies between North, 360°/000°, and West, 270°, closer to North-West, 315°)

24. 160° (340° − 180° = 160°)

25. The back bearing points roughly North; adding 180° to 162° gives a back bearing of 342°

26. C) 185° (005° + 180° = 185°)

27. A, C, D and E are true (B is false — since 205° is 180° or more, the back bearing is found by SUBTRACTING 180°, not adding it)

28. 145° (325° − 180° = 145°)

29. 135° (South-East is a bearing of 135°)

30. 268° (088° + 180° = 268°)

Standard / Advanced

31. 276° (096° + 180° = 276°)

32. 025° (205° − 180° = 025°)

33. 120° (170° − 050° = 120°)

34. C) 200° (110° + 90° = 200°)

35. Corrected bearing = 182° (162° + 20°); back bearing = 002° (182° − 180°)

36. A, B, C and E are true (D is false — if the shop is south-west of the house, then the house is north-east of the shop, not south-east)

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37. 90° (the angle between bearings 040° and 310°, measured the short way round, is 040° + (360° − 310°) = 040° + 50° = 90°)

38. A) They are the same point (115° + 180° = 295°, which exactly matches the B-to-C bearing, so the second leg retraces the first leg in reverse)

39. 248° (068° + 180° = 248°)

40. A, B, D and E are true (C is false — 180° − 134° = 46°, which is not the back bearing; the correct back bearing is 134° + 180° = 314°)

41. 136° (316° − 180° = 136°); this is closest to C) South-East (135°)

42. 018° (198° − 180° = 018°); this is closest to North (000°)

43. Since 292° is exactly the back bearing of 112° (112° + 180° = 292°), the B-to-C leg lies on the same straight line as the A-to-B leg, heading back towards A. As B-to-C is 4 km (shorter than the 9 km A-to-B leg), C is 9 − 4 = 5 km from A. Direct distance A to C = 5 km; via B the ship travelled 9 + 4 = 13 km; so sailing directly would have saved 13 − 5 = 8 km

44. $x + 210 = 4x + 180$, so $210 - 180 = 4x - x$, giving $x = 10$ (outward bearing = 4(10) = 40°, back bearing = 10 + 210 = 220°, and 40° + 180° = 220° ✓)

45. $3y + 10 = (y + 20) + 180$, so $3y + 10 = y + 200$, giving $2y = 190$, so $y = 95$ (outward bearing = 95 + 20 = 115°, back bearing = 3(95) + 10 = 295°, and 115° + 180° = 295° ✓)

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